Hierarchical information-gating diagnostic

Study 7 shows that the 3-level agent runs and federates end to end. This supplementary diagnostic asks a narrower question: under a configured surprise threshold,

Hierarchical... (part 2)

does one leaf observation classify the top meta-context as informative or non-gating? The output is a thresholded information-gating diagnostic,

Hierarchical... (part 3)

not a Bayesian-model-evidence comparison or a data-driven proof of the hierarchy's correct depth.

Hierarchical… (part 4)

The implementation evaluates levels through `hierarchical_reduce` in the `bayesian_model_reduction` module of `fedference`. For each non-leaf level we measure its **Bayesian surprise** $\mathrm{KL}(q_i \,\|\, \tilde{p}_i)$ —

Hierarchical... (part 5)

how far the leaf observation moves that level's belief q_i from its top-down prior $\tilde{p}_i$.

Hierarchical… (part 6)

The routine labels a level below the configured threshold as prunable and a level above it as kept. Because this is an inference-derived divergence rather than a model re-fit, the labels are conditional on the specified worlds, observation, and threshold.

Hierarchical… (part 7)

The control is directional by construction. We build two 3-level worlds that differ *only* in the top level's conditioned priors:

Hierarchical... (part 8)

a **degenerate** world whose meta-context is non-gating (both meta-context states predict the same context distribution) and an **informative** world whose meta-context sharply distinguishes the two contexts.

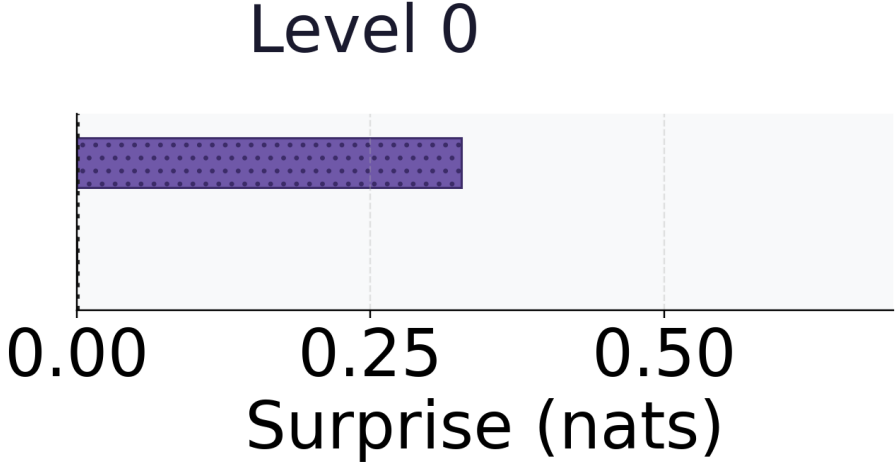
Hierarchical... (part 9)

In the degenerate world, the top level has Bayesian surprise 0.000 nats and is flagged prunable (recovers the two-level structure: Yes). In the informative world, it has 0.328 nats and is kept (Yes).

Hierarchical... (part 10)

fig. 35 shows both configured worlds side by side. Their shared lower-level parameters isolate the top-level gating contrast, but do not establish a generally correct depth-selection procedure.

Hierarchical... (part 11)



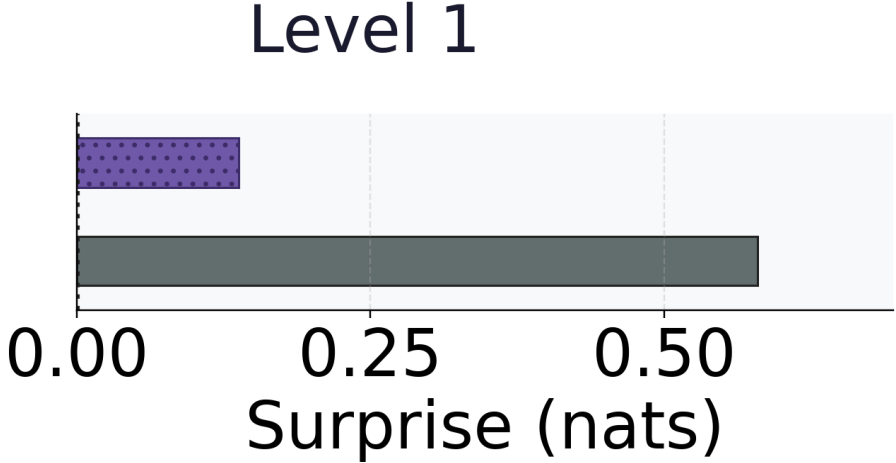
Hierarchical... (part 12)

non-gating world: L0 low_threat. Bayesian surprise 0 nats; source flag: prunable.

Hierarchical… (part 13)

informative world: L0 low_threat.
Bayesian surprise 0.3277 nats;
source flag: kept.

Hierarchical... (part 14)



Hierarchical… (part 15)

non-gating world: L1 quiet. Bayesian surprise 0.5796 nats; source flag: kept.

Hierarchical… (part 16)

informative world: L1 quiet. Bayesian surprise 0.138 nats; source flag: kept.

Hierarchical… (part 17)

Declared surprise threshold: 0.001 nats. prunable iff bayesian_surprise < surprise_tol; recommend the deepest prunable non-leaf level

Hierarchical... (part 18)

Lower bar: non-gating configured world. Upper hatched bar: informative configured world. All levels share the same surprise scale.

Hierarchical... (part 19)

Bayesian surprise KL(posterior || empirical prior). This configured information-gating diagnostic has no stochastic interval and is distinct from exact BMR evidence gain.

Hierarchical... (part 20)

Unlike the Beta-function BMR sign control in sec. 10.4 and eq. 20, this routine thresholds per-level surprise. It therefore checks the intended gating contrast in the declared worlds; it does not let the data independently choose model depth.